(i)

Classification	No of Std	proportion	Sample size (prop x n)
B.Sc	150	150/728	$\frac{150}{728} \times 40 = 8$
B.A	163	163/728	$\frac{163}{728} \times 40 = 9$
F.Sc	195	195/728	$\frac{195}{728} \times 40 = 11$
F.A	220	220/728	$\frac{220}{728} \times 40 = 12$
	728		40

Final results

The sample size for each classification are

B.Sc : 8 students

B.A : 9 students

F.Sc : 11 Students

F.A : 12 Students

A.R

(ii) Given:

$n = 100$

$\hat{p} = \frac{\text{Number of success}}{\text{Total sample size}} = \frac{16}{75} = 0.2133$

$z = 1.645$ (from z table for 90% confidence interval)

Calculate standard error

$S.E = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{0.2133(1-0.2133)}{75}}$

$= \sqrt{\frac{0.2133 \times 0.7867}{75}} = \sqrt{0.002236}$

$= 0.0473$

M.E (margin of error) $= Z \times S.E = 1.645 \times 0.0473$

$= 0.0778$

$C.I = \hat{p} \pm ME = 0.2133 \pm 0.0778$

$C.I = 0.2133 + 0.0778$	$C.I = 0.2133 - 0.0778$
$C.I = 0.2911$	$C.I = 0.1355$

(iii) Step 1. (Test hypothesis):

$H_0 : \mu \geq 50$

$H_a : \mu < 50$

Since sample size is 36 (large enough) we use z test for test statistics

$$z = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Here $\bar{x} = 48.7$

$\mu_0 = 50$

$s = 3.1$

$n = 36$

Butting values

$$z = \frac{48.7 - 50}{3.1/\sqrt{36}} = \frac{-1.3}{3.1/6}$$

$$z = \frac{-1.3}{0.5167}$$

$$z = -2.52$$

Here $\alpha = 0.05$ (typical threhold for rejection ~~for~~ in hypothesis test)

$z_{critical (0.05)/2} = 0.975$

z critical $= -1.645$

The calculated z-value is -2.52

critical value is -1.645

$-2.52 < -1.645$

The test hypothesis falls ~~for~~ in the rejection region.

Reject the null hypothesis. there is a significant evidence to conclude that average age of ~~(student)~~ members of parliaments is less than 50 years.

(iv) Step 1:-

$$H_0 : \mu_2 - \mu_1 \le 10$$

$$H_a : \mu_2 - \mu_1 > 10$$

Given:

$$n_1 = 12, \bar{x}_1 = 10, S_1^2 = 1200$$

$$S_1 = \sqrt{1200} = 34.64$$

$$n_2 = 18, x_2 = 25, S_2^2 = 900$$

$$S_2 = \sqrt{900} = 30, \quad \Delta_0 = 10$$

Significance level $= \alpha = 0.05$

$$t = \frac{(\bar{x}_2 - \bar{x}_1) - \Delta_0}{\sqrt{\frac{S_1^2}{n} + \frac{S_2^2}{n_2}}}$$

$$t = \frac{(15-10)}{\sqrt{\frac{1200}{15} + \frac{900}{180}}}$$

$$= \frac{5}{\sqrt{100+50}} = \frac{5}{\sqrt{150}} = \frac{5}{12.247}$$

$t = 0.408$

Determine the degree of freedom

$$df = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\left(\frac{S_1^2}{n_1}\right)^2 + \left(\frac{S_2^2}{n_2}\right)^2}$$

$$= \frac{\left(\frac{1200}{12} + \frac{900}{18}\right)^2}{\frac{(1200/12)^2}{11} + \frac{(900/18)^2}{17}}$$

$$= \frac{(100+50)^2}{\frac{10000}{11} + \frac{2500}{17}}$$

$$= \frac{22500}{909.09 + 147.06}$$

$$df = \frac{22500}{1056.15} = 21.3$$

$df = 21$

- for critical angle
  $\alpha = 0.05$ and $df = 21$
  $t_{critical\,(0.05)(2)} = 1.721$

- Compare tvalue with tcritical

  $t = 0.408 < 1.721$
  we fail to reject the null hypothesis
  Therefore $\alpha = 0.05$ level to support the claim that
  $\mu_2 - \mu_1 \geq 10$.

(v) Step 1:-
$H_0: \sigma^2 \leq 7.7953$
$H_a: \sigma^2 > 7.753$

Sample(x)	$(x - \bar{x})$	$(x - \bar{x})^2$
67.50	-0.15	6.025
70.15	3.1	9.9225
72.00	4.35	18.9225
63.25	-4.4	19.6025

x	$(x-\bar{x})$	$(x-\bar{x})^2$
65.25	-2.4	5.6225
68.75	1.1	1.1025
69.50	1.85	2.7255
68.50	0.85	0.7225
66.50	-1.15	1.3225
64.75	-2.9	8.4025
676.50		68.36

$$\bar{x} = \frac{\Sigma x}{n} = \frac{676.50}{10}$$

$$\bar{x} = 67.65$$

$$S^2 = \frac{\Sigma(x-\bar{x})^2}{n-1} = \frac{68.36}{10-1}$$

$$S^2 = \frac{68.36}{9} = 7.596$$

The test statistics for variance uses the chi-square distribution

$$\chi^2 = \frac{(n-1)\,s^2}{\sigma_0^2}$$

$$= \frac{(10-1)(7.596)}{7.7953} = \frac{9 \times 7.596}{7.7953}$$

$$= \frac{68.364}{7.7953}$$

$\chi^2_{(0.01),(9)} = 21.666$

Test Statistics $(\chi^2) = 8.77$

Critical value $= 21.666$

$\chi^2 < \chi_{critical}$

$8.77 < 21.666$

We fail to reject the null hypothesis. There is insufficient evidence at the $\alpha = 0.01$ level to conclude that $\sigma > 7.7953$

10-12-2024